Gaurav Chintakunta

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Summary

Software engineer with an MS in CS from UIC (GPA 3.89), currently modernizing a legacy scientific computing toolkit as a Research Scientist. Builds Python services, AI agents, and data pipelines with tests and CI on every change, and has 5 merged open source PRs fixing bugs in large codebases.

Education

University of Illinois at Chicago | MS in Computer Science | GPA - 3.89/4.0 *May 2026*
Coursework: Distributed Systems, Object-Oriented Design, Neural Networks, Responsible AI Engineering

JNTU Hyderabad | Bachelor of Technology in Computer Science *May 2024*
Coursework: Design and Analysis of Algorithms, Data Structures, Software Engineering, Machine Learning

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Go, Fortran
- **Frameworks and Data:** FastAPI, React, Next.js, pandas, NumPy, TensorFlow, PyTorch, PostgreSQL
- **AI and Agents:** OpenAI API, Claude, PydanticAI, LangChain, CrewAI, MCP, RAG, LLM Evaluation
- **Infrastructure:** Docker, Kubernetes, AWS, Linux, Prometheus, GitHub Actions CI/CD
- **Practices:** Legacy Code Modernization, Regression Testing, Monte Carlo and Latin Hypercube Sampling, Code Review

Experience

- **Research Scientist** *Sep 2026 - Present*
  University of Illinois at Chicago Remote
  - Mapped the full architecture of a 7,500-line Fortran engine and 34-form VB6 GUI by reverse-engineering the undocumented code.
  - Surfaced latent bugs, like keywords that silently truncate input, by auditing the Fortran parser line by line against the manuals.
  - Won PI approval for a modernization plan by designing Python orchestration around an isolated Fortran subprocess.
  - Safeguarding numerical parity with the legacy system by building regression tests and Python validation for 17+ distributions.
- **Open Source Contributor** *Apr 2026 - Present*
  5 merged PRs across Hugging Face, deepset, and other open source AI frameworks Remote
  - **Haystack (26K+ stars):** Fixed a silent ranking bug in code I had never seen, where one empty metadata value disabled sorting for every downstream user, with tests for every missing-data case (PR #12863, merged).
  - **Sentence Transformers by Hugging Face (19K+ stars):** Fixed hard-negative mining silently skipping the reranking model when only a minimum score was set, unifying the filter logic and adding a regression test (PR #4073, merged).
- **Graduate Researcher** *Jan 2025 - Jan 2026*
  University of Illinois at Chicago, Computer Science Chicago, IL
  - Boosted prediction accuracy by 30% with Pix2Path, a TensorFlow GAN predicting Alzheimer's plaques from spatial gene data.
  - Built the pipeline turning raw Xenium transcripts into 534-channel gene images with aligned, thresholded plaque labels.
  - Accelerated training by 40% per epoch by profiling the pipeline on UIC's HPC cluster and speeding up data loading.
  - Stabilized long GAN runs by pairing adversarial and L1 losses, checkpointing often, and tracking losses in TensorBoard.

Projects

- **Winnow: AI Inbox Triage Platform | Python, FastAPI, PostgreSQL, PydanticAI, Next.js, Docker** GitHub
  - Matched full-LLM routing at classifier speed and near-zero cost by sending only low-confidence email to an LLM agent.
  - Improved the model from real usage by retraining nightly on user corrections, gated by score checks with one-command rollback.
- **TraceGuard: Agent Run Tracing and Review | TypeScript, Next.js, Cloudflare Workers, D1, OpenAI API** GitHub
  - Accelerated agent debugging by tracing every step, tool call, failure, cost, and latency through a REST ingestion API.
  - Eliminated model-made policy decisions by moving them into deterministic code, so the LLM only drafts and a human approves.
- **GridScout: Battery Storage Site Intelligence | Python, GeoPandas, Streamlit, Folium, OpenAI API** GitHub
  - Surfaced promising Texas storage sites by cleaning HIFLD, EIA-860, and Census data into meter-based spatial features.
  - Ensured every ranking was auditable by explaining each score on a live map and validating every citation in LLM memos.
- **Autonomous Agentic SRE Pipeline | Python, Go, CrewAI, MCP, Kubernetes, Prometheus** GitHub
  - Automated incident response with agents that diagnose Prometheus alerts and act on Kubernetes through a custom MCP server.
  - Contained risky automation by restricting agents to allowlisted tools and escalating only when a simple restart failed.